ABGs

Arterial Blood Gases

Respiratory Acidosis

$pH < 7.35$ $PaCO_2 > 45$

Respiratory **O**pposite

($pH + CO_2$ move in opposite direction ↓↑)

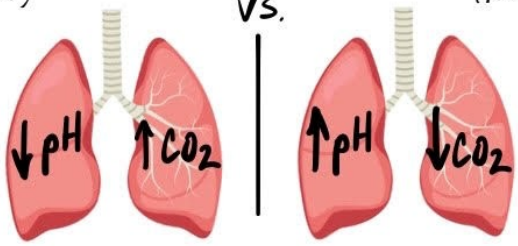
vs.

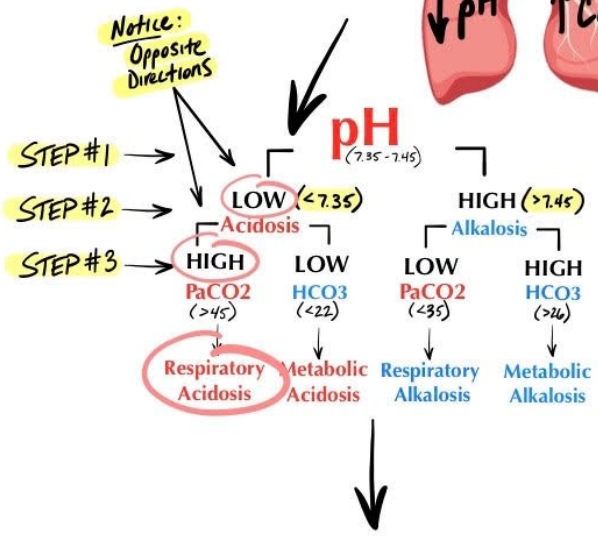
Respiratory Alkalosis

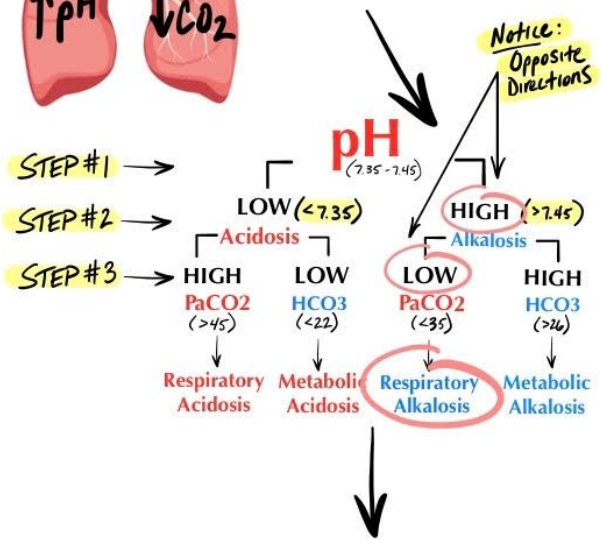
$pH > 7.45$ $PaCO_2 < 35$

Respiratory **O**pposite

($pH + CO_2$ move in opposite direction ↓↑)







Causes

- Hypoventilation
- Respiratory Depression (i.e. Narcotics, Head injury)
- Respiratory muscle weakness
- Asthma
- COPD
- Pneumonia
- Pulmonary Edema
- Drowning

Think: conditions that would impair respiration + the ability to exhale CO_2

Signs/Symptoms

- Bradycardia
- Hypotension
- Agitation
- Confusion
- Combativeness
- Lethargy
- Coma

Think: altered mental status, trouble staying awake, retaining CO_2

Treatments

- Reposition the patient
- Maintain patent airway
- Ventilation as needed to remove excess CO_2

Causes

- Hyperventilation
- Hypermetabolic states
  - Fever
  - Anemia
  - Thyrotoxicosis
  - Septicemia
- Anxiety
- Fear
- Pain
- Pneumothorax

Think: conditions causing respiration increase (tachypnea) leading to exhaling too much CO_2 (acidic)

Signs/Symptoms

- Tachycardia
- Anxiety
- Confusion
- Diaphoresis
- Dizziness
- Coma

Treatments

- Control breathing
- Sedatives PRN
- Rebreather mask
- Correct for metabolic causes